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Serial.begin()

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Description

Sets the data rate in bits per second (baud) for serial data transmission. When communicating with the Serial Monitor, ensure that you use one of the baud rates listed in the Serial Monitor's baud rate dropdown menu. You can, however, specify other rates—for example, to communicate over pins 0 and 1 with a component that requires a particular baud rate.

An optional second argument configures the data, parity, and stop bits. The default is 8 data bits, no parity, one stop bit.

Syntax

Use the following function to initialize and configure the serial communication:

`Serial.begin(baud)`

`Serial.begin(baud, config)`

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Parameters

The function admits the following objects and parameters:

`Serial`: serial port object. See the list of available serial ports for each board on the [Serial main page](#).

`baud`: in bits per second (baud). Allowed data types: `long`.

`config`: sets data, parity, and stop bits. Valid values are:

`SERIAL_5N1`

`SERIAL_6N1`

`SERIAL_7N1`

`SERIAL_8N1` (the default)

`SERIAL_5N2`

`SERIAL_6N2`

`SERIAL_7N2`

`SERIAL_8N2`

`SERIAL_5E1`: even parity

`SERIAL_6E1`

`SERIAL_7E1`

`SERIAL_8E1`

`SERIAL_5E2`

`SERIAL_6E2`

`SERIAL_7E2`

`SERIAL_8E2`

`SERIAL_5O1`: odd parity

`SERIAL_6O1`

`SERIAL_7O1`

`SERIAL_8O1`

`SERIAL_5O2`

`SERIAL_6O2`

`SERIAL_7O2`

`SERIAL_8O2`

Returns

RETURNS

The function returns **nothing**.

Example Code

```
1 void setup() {  
2     Serial.begin(9600)  
3 }  
4  
5 void loop() {}
```

Multiple serial example:

```
1 // Arduino Mega using  
2 // (Serial, Serial1,  
3 // with different bau  
4  
5 void setup() {  
6     Serial.begin(9600  
7     Serial1.begin(384  
8     Serial2.begin(192  
9     Serial3.begin(480  
10  
11     Serial.println("H  
12     Serial1.println("  
13     Serial2.println("  
14     Serial3.println("  
15 }  
16 void loop() {}
```

Thanks to Jeff Gray for the mega example.

Notes and Warnings

For USB CDC serial ports (e.g.

`Serial` on the Leonardo),

`Serial.begin()` is irrelevant.

You can use any baud rate and

See the list of available serial
ports for each board on the
[Serial main page](#).

The only `config` value
supported for `Serial1` on the
Arduino Nano 33 BLE and Nano
33 BLE Sense boards is
`SERIAL_8N1`.

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